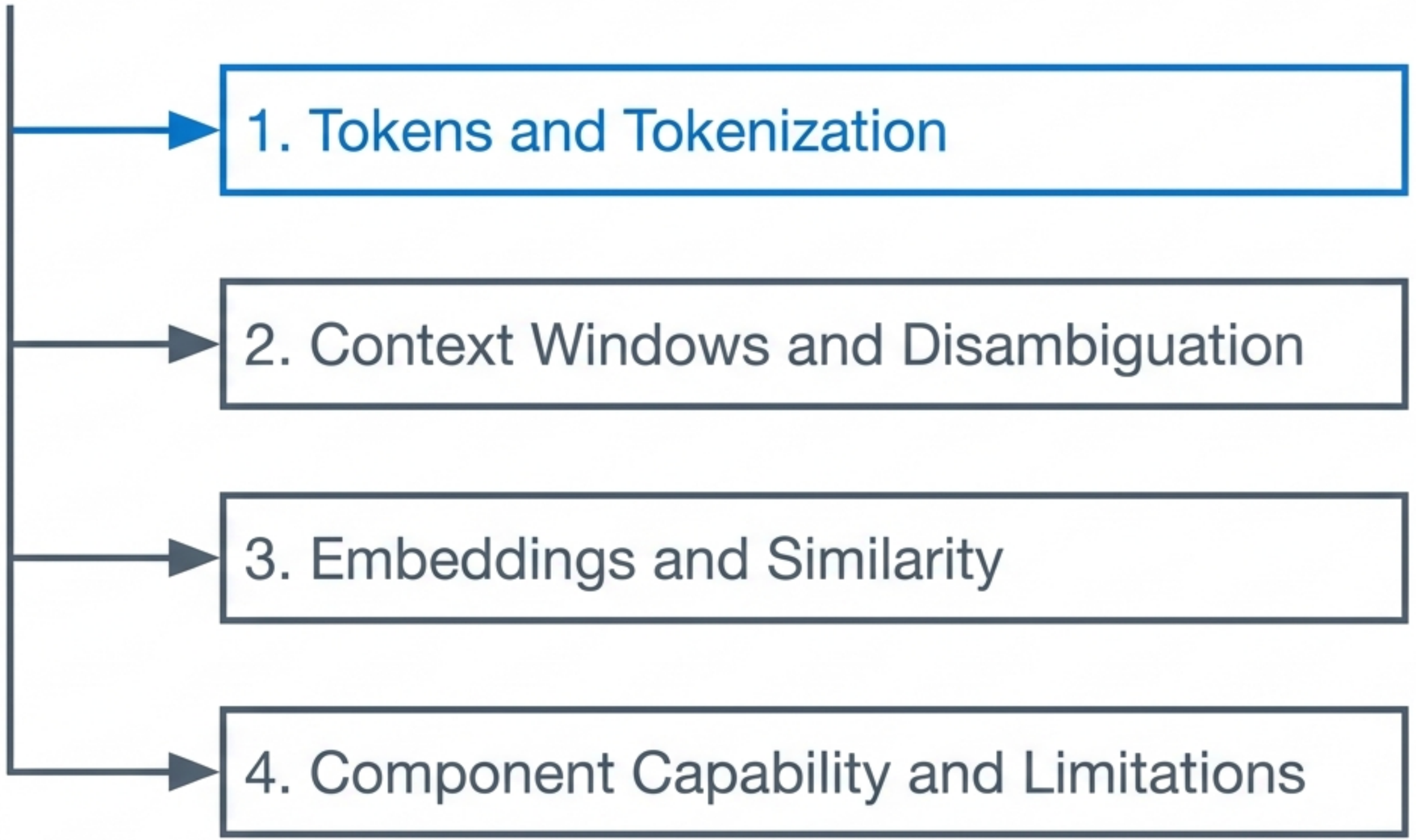
Module 13, Lesson 2

AI 101 for Mechanical Engineers: Tokens, context and embeddings

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System Pipeline Outline



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1. Tokens and Tokenization

Purpose: Define the discrete encoded units processed by models.

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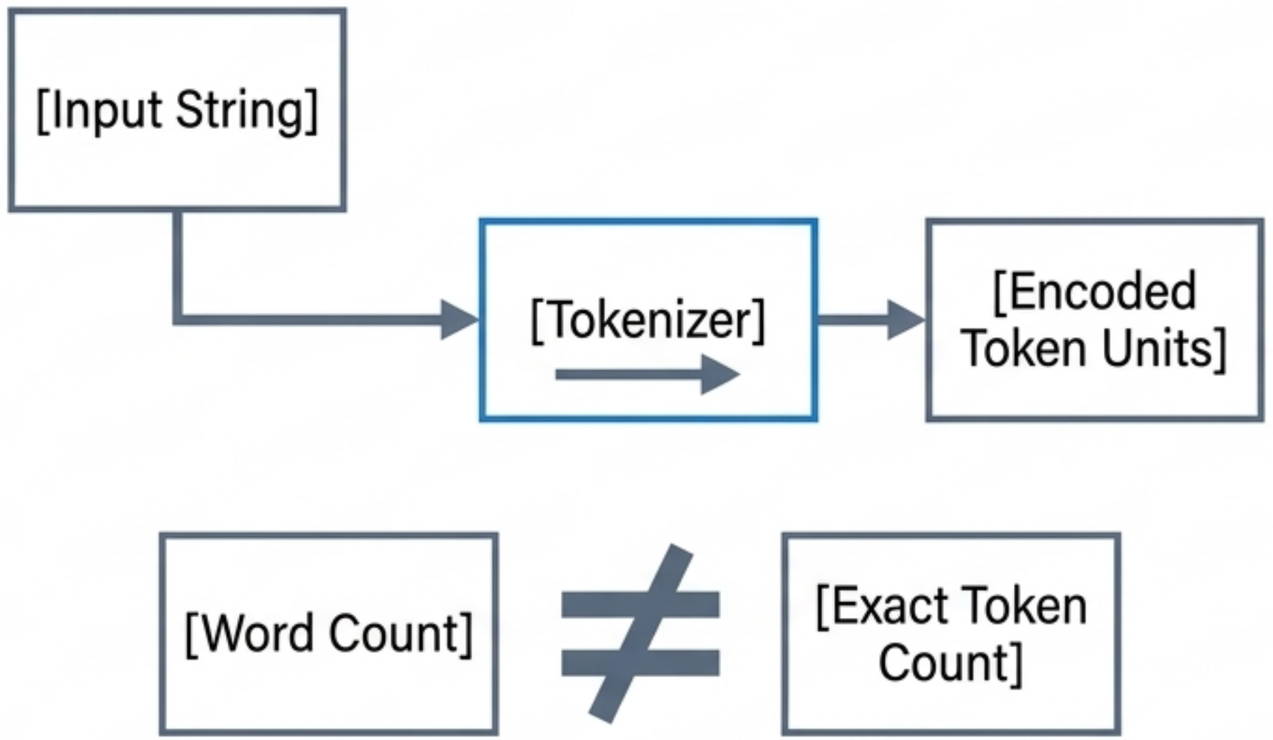
Tokenization Depends on Specific Model Architecture

Tokens represent discrete, model-specific encoded processing units.

Units may constitute word fragments, whole words, or punctuation.

Individual model architecture defines exact tokenization rules and boundaries.

Human word counts do not equate to exact machine token counts.



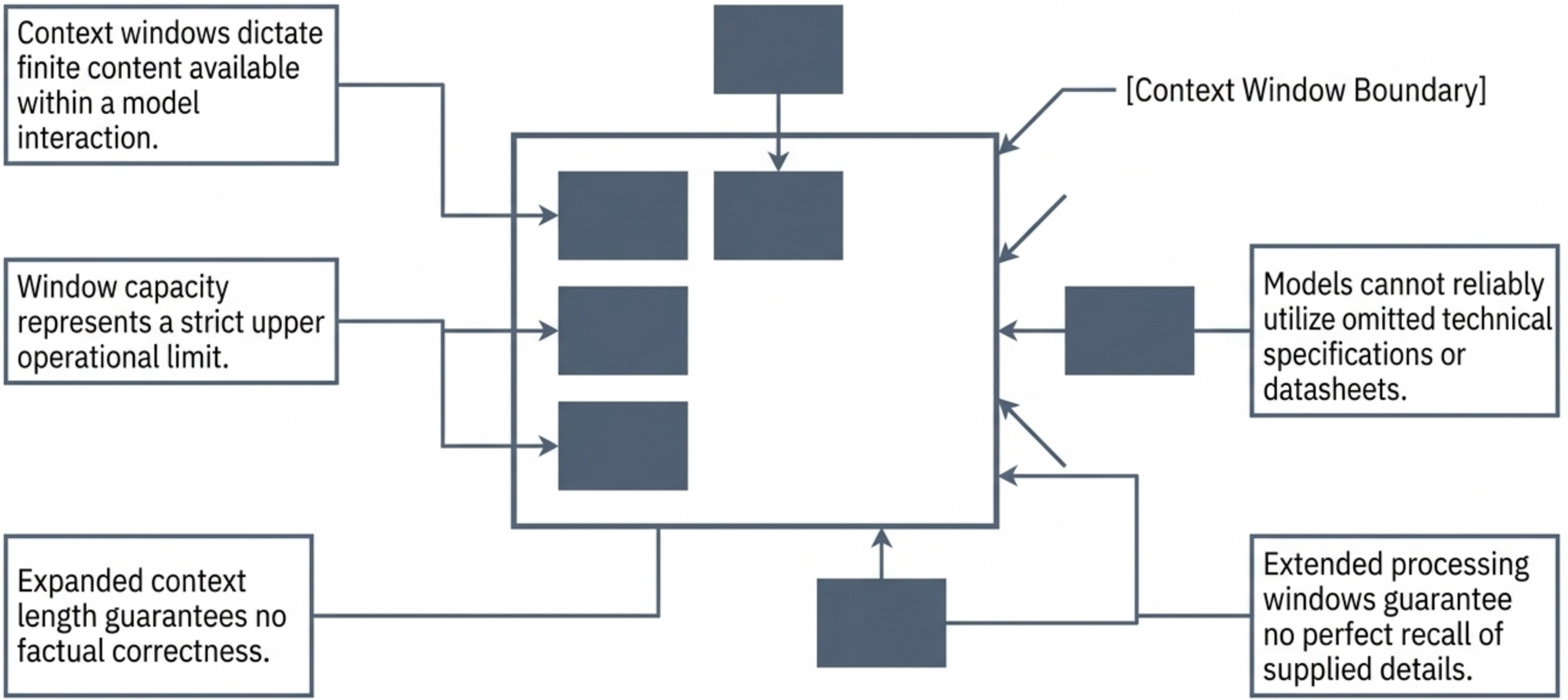
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2. Context Windows and Disambiguation

Purpose: Establish the operational interaction boundaries and attention mechanisms.

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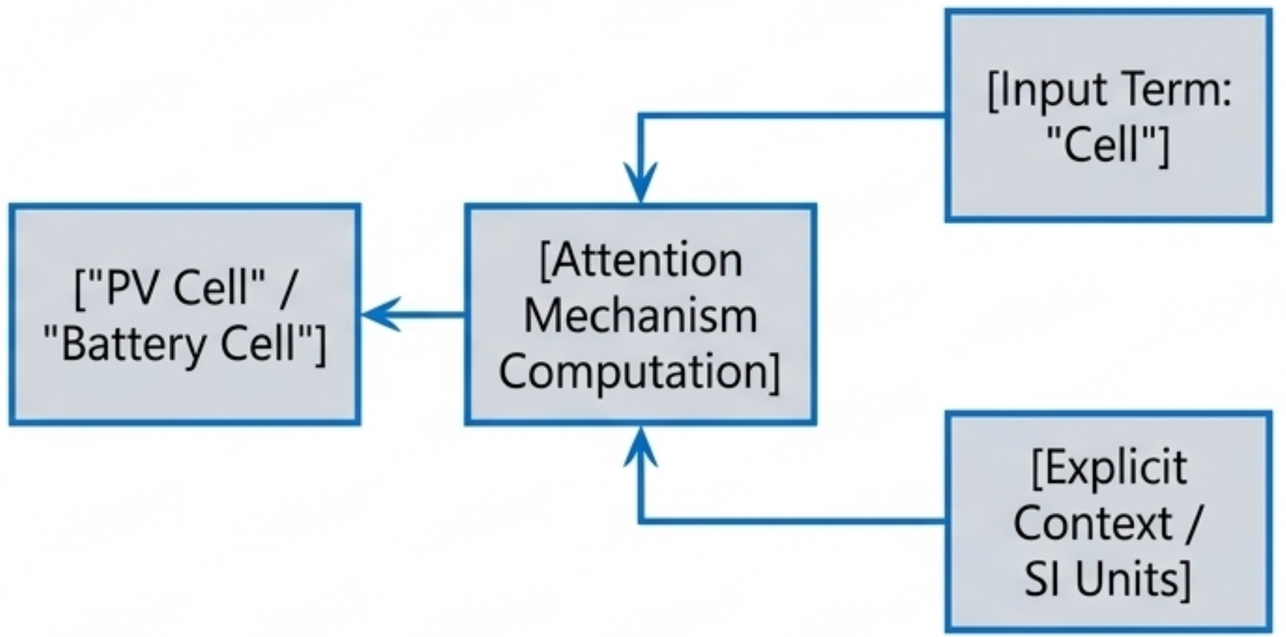
Finite Content Processing Boundaries



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Attention Mechanisms Require Explicit System Parameters

- Attention mechanisms evaluate representations via learned mathematical computations.
- Ambiguous terminology requires explicitly defined contextual vectors.
- Operators must explicitly specify target mechanical systems.
- Operators must supply exact SI units for all numerical parameters.
- Models cannot safely infer missing operational assumptions.



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3. Embeddings and Similarity

Purpose: Detail the mathematical vector representation of discrete items.

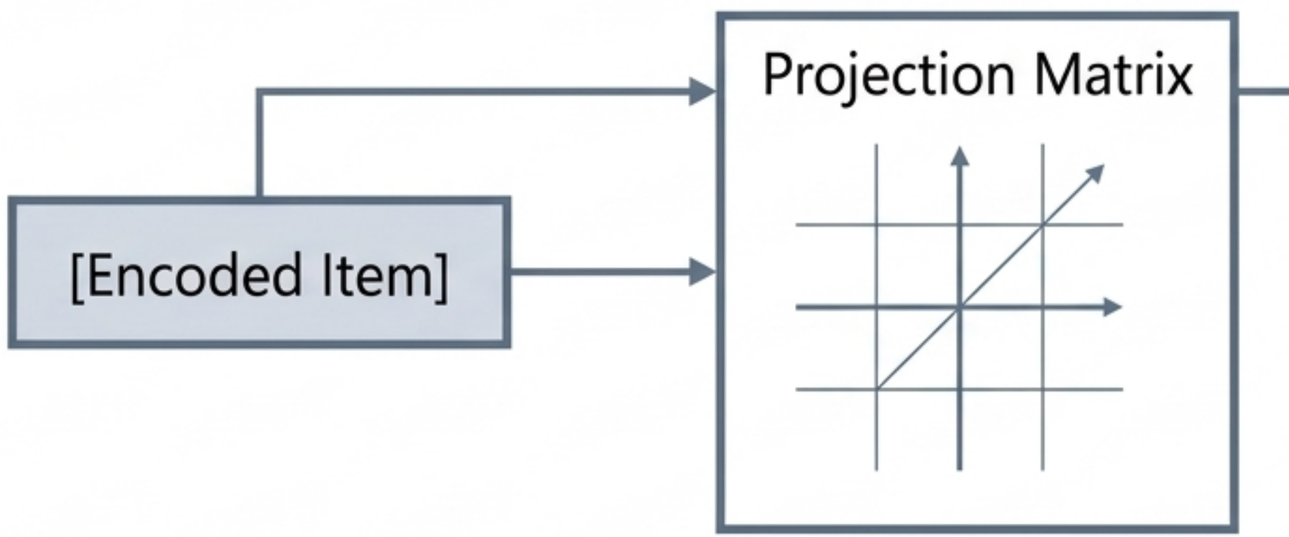
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Mathematical Vector Representation Array

Systems convert discrete textual items into formal mathematical embeddings.

Embeddings consist exclusively of numerical vector representations.

This mathematical formatting enables automated computational processing and comparison.



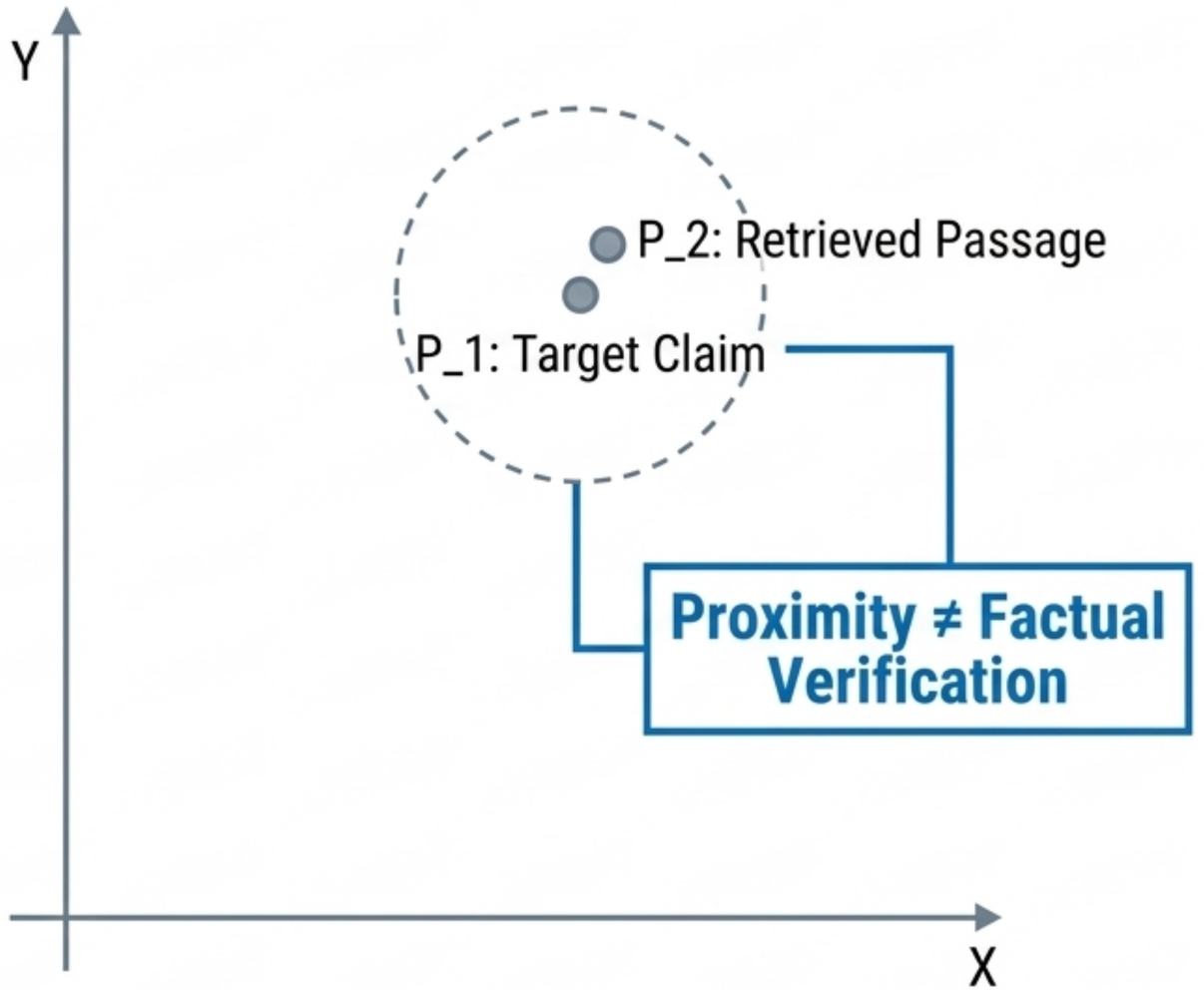
$$\mathbf{E} = [\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_n}]$$

(where variables represent continuous numerical coordinates)

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Vector Proximity Does Not Constitute Verification

- Vector similarity facilitates the retrieval of mathematically related data.
- Proximity identifies related reference material, such as manual passages.
- Vector distance determines mathematical relatedness between terms.
- Relatedness metrics provide zero proof of operational claim validity.
- Similarity algorithms cannot verify factual correctness.



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4. Component Capability and Limitations

Purpose: Synthesize module components into operational engineering constraints.

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Component Capability and Limitation Matrix

System Component	Operational Function	Engineering Limitation
Token	Processes discrete encoded model-specific units.	Exact token counts cannot be mapped reliably from word counts.
Context Window	Bounds available information during model interaction.	Provides no guarantee of perfect detail recall or correct usage.
Embedding	Projects encoded items as numerical vectors.	Operates purely as a mathematical representation array.
Similarity	Retrieves mathematically related baseline passages.	Yields relatedness data without proving claim factual support.

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Mandatory Independent Operator Verification

[AI System Output] → **[Independent Operator Verification]**

- Similarity algorithms cannot validate real equipment safety specifications.
- Long context processing cannot guarantee omission-free outputs.
- Engineers must independently supply omitted technical datasheets directly.
- Engineers must explicitly state all operational assumptions.
- Engineers must independently verify output factual correctness.

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Key Points

1	1. Model-specific token limits govern inputs, superseding human word counts.
2	2. Bounded context windows require explicit parameters and SI unit declarations.
3	3. Embeddings project textual items into mathematical vector arrays for retrieval.
4	4. Vector similarity and long context windows yield relatedness without guaranteeing factual verification.

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